

MATH403: Homework 4

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10. Write $(13)(1245)(13)$ and $(24)(13456)(24)$ in disjoint cycle form. Give a simple description of how each product cycle compares with middle cycle in each product.

Proof.

1. Denote this permutation α .

- $\alpha(1) \implies (13)$ sends 1 to 3, then (1245) fixes 3, (13) sends 3 back to 1.
- $\alpha(2) \implies (13)$ fixes 2, (1245) sends 2 to 4, then (13) fixes 4.
- $\alpha(3) \implies (13)$ sends 3 to 1, (1245) sends 1 to 2, and (13) fixes 2.
- $\alpha(4) \implies (13)$ fixes 4, (1245) sends 4 to 5, then (13) fixes 5.
- $\alpha(5) \implies (13)$ fixes 5, (1245) sends 5 to 1, then (13) sends 1 to 3.

This yields the disjoint cycle form $(1)(2453)$. Compared to the middle cycle $(1245) = (1245)(3)$, the product cycle essentially makes 1 the identity cycle, and replaces 1 with 3 in the middle cycle.

2. Denote this permutation β .

- $\beta(1) \implies (24)$ fixes 1, (13456) sends 1 to 3, and (24) fixes 3.
- $\beta(2) \implies (24)$ sends 2 to 4, (13456) sends 4 to 5, and (24) fixes 5.
- $\beta(3) \implies (24)$ fixes 3, (13456) sends 3 to 4, and (24) sends 4 to 2.
- $\beta(4) \implies (24)$ sends 4 to 2, (13456) fixes 2, and (24) sends 2 to 4.
- $\beta(5) \implies (24)$ fixes 5, (13456) sends 5 to 6, and (24) fixes 6.
- $\beta(6) \implies (24)$ fixes 6, (13456) sends 6 to 1, and (24) fixes 1.

This yields the disjoint cycle $(13256)(4)$. Compared to the middle cycle $(13456) = (13456)(2)$, the product cycle essentially makes (4) the identity cycle, and replaces 4 with 2 in the middle cycle.

□

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30. Let α and β belong to S_n . Prove that $\alpha^{-1}\beta^{-1}\alpha\beta$ is an even permutation.

Proof. By **Theorem 5.4** in Gallian, every permutation in S_n where $n > 1$ can be written as a product of two cycles. Without loss of generality, α omits the decomposition,

$$\alpha = (a_1 a_k)(a_1 a_{k-1}) \cdots (a_1 a_2)(b_1 b_t)(b_1 b_{t-1}) \cdots (b_1 b_2) \cdots (c_1 c_s)(c_1 c_{s-1}) \cdots (c_1 c_2)$$

Let k be the number of 2-cycles to represent the permutation α . By **Theorem 5.5**, k is either always even or always odd. Similarly, let r be the number of 2-cycles to represent β , which is also always even or odd. Note that if a permutation is even or odd, then the inverse of the permutation is the also the same parity. Now let us take cases to show that irrespective of the parity of α and β , the product permutation $\alpha^{-1}\beta^{-1}\alpha\beta$ is always an even permutation. To do this, we will just sum the number of 2-cycles in the product; since **Theorem 5.5** holds, if this sum is even, then we can immediately conclude that the product permutation is even.

1. α, β are both even $\implies$ even + even + even + even $\implies$ even
2. α even and β odd $\implies$ even + odd + even + odd $\implies$ even
3. α odd and β even $\implies$ odd + even + odd + even $\implies$ even
4. α, β both odd $\implies$ odd + odd + odd + odd $\implies$ even

Therefore, the product cycle $\alpha^{-1}\beta^{-1}\alpha\beta$ is an even permutation, as desired. $\square$

40. Let $\beta = (123)(145)$. Write β^{99} in disjoint cycle form.

Proof. Given that $\beta^1 = (123)(145)$, note that $1 \rightarrow 4, 2 \rightarrow 3, 3 \rightarrow 1, 4 \rightarrow 5, 5 \rightarrow 2, 6 \rightarrow 6$ and so the disjoint cycle form is

$$\beta^1 = (123)(145) \implies (14523)$$

This cycle is of order 5, so therefore, $\beta^1 = \beta^6$. Note that $99 \equiv 4 \pmod{5}$. Therefore,

$$\beta^{99} = \beta^4 = (14523)(14523)(14523)(14523)$$

and since $1 \rightarrow 3, 2 \rightarrow 5, 3 \rightarrow 2, 4 \rightarrow 1, 5 \rightarrow 4$, and this implies that

$$\beta^{99} = (13254)$$

□

74. (Indiana College Mathematics Competition) A card-shuffling machine always rearranges cards in the same way relative to the order in which they were given to it. All of the hearts arranged in order from ace to king were put into the machine, and then the shuffled cards were put into the machine again to be shuffled. If the cards emerged in the order 10, 9, Q, 8, K, 3, 4, A, 5, J, 6, 2, 7, in what order were the cards after the first shuffle?

Proof. Let us denote the rank of the card numerically such that $A = 1, 2 = 2 \cdots, K = 13$. We seek a permutation

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 \\ \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 & \alpha_5 & \alpha_6 & \alpha_7 & \alpha_8 & \alpha_9 & \alpha_{10} & \alpha_{11} & \alpha_{12} & \alpha_{13} \end{bmatrix}$$

such that

$$\alpha^2 = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 \\ 10 & 9 & 12 & 8 & 13 & 3 & 4 & 1 & 5 & 11 & 6 & 2 & 7 \end{bmatrix}$$

and from here, the sequence $\alpha_1, \dots, \alpha_{13}$ will be the order of the cards after the first shuffle. Note, however, that since the card-shuffling machine always rearranges card in the same relative order, α^{13} will be the identity. Thus, to find the order after the first shuffle, we need only find the order after the 14th shuffle, and this will be the same order. To do this, we must apply α^2 six more times. To help with this, I wrote a simple Python script to find the order after $\alpha^{14} = \alpha^1$.

```

1 alpha_squared = {1: 10, 2: 9, 3: 12, 4: 8, 5: 13, 6: 3, 7: 4, 8: 1, 9: 5,
2   10: 11, 11: 6, 12: 2, 13: 7}
3 curr = {i: i for i in range(1, 14)}
4 for i in range(7):
5     curr = {j: curr[alpha_squared[j]] for j in range(1, 14)}
6 print(curr)

```

The result of `curr` is {1: 9, 2: 1, 3: 4, 4: 12, 5: 11, 6: 7, 7: 3, 8: 2, 9: 10, 10: 5, 11: 13, 12: 8, 13: 6}. Converting this back to our cards ranks, we obtain that the original order of the cards is

9, A, 4, Q, J, 7, 3, 2, 10, 5, K, 8, 6

□

30. Prove that every automorphism α of $\mathbf{R}^*$, the group of real numbers under multiplication, $\alpha(-1) = -1$.

Proof. Let $\alpha : \mathbf{R}^* \rightarrow \mathbf{R}^*$ be a mapping such that we have an automorphism of $\mathbf{R}^*$ — then it follows that α is 1-1, onto, and operation preserving. Furthermore, by **Property 1 of Theorem 6.1**, α carries the identity of $\mathbf{R}^*$ to the identity of $\mathbf{R}^*$. Therefore, we have

$$\alpha(1) = 1$$

Since $-1 \in \mathbf{R}^*$, we can apply the automorphism

$$1 = \alpha(1) = \alpha(-1 * -1) = \alpha(-1)\alpha(-1)$$

which implies that $\alpha(-1) = -1, 1$. However, α is 1-1, and therefore,

$$\alpha(-1) = \alpha(1) \implies -1 = 1$$

which is a contradiction. Therefore, $\alpha(-1) = -1$ must be true to satisfy the definition of automorphism, as desired. $\square$

48. Prove that $\mathbf{R}$ under addition is not isomorphic to $\mathbf{R}^*$ under multiplication.

Proof. Assume on the contrary that there exists a function $\phi : \mathbf{R} \rightarrow \mathbf{R}^*$ such that $\mathbf{R}$ under addition is isomorphic to $\mathbf{R}^*$ under multiplication. Then, it follows that ϕ is both 1-1, onto, and operation-preserving. If ϕ were such a mapping, there would exist a real number $x \in \mathbf{R}$ such that $\phi(x) = -1$ as $-1 \in \mathbf{R}^*$. However, observe that

$$-1 = \phi(x) = \phi\left(\frac{1}{2}x + \frac{1}{2}x\right) = \phi\left(\frac{1}{2}x\right) \phi\left(\frac{1}{2}x\right) = \left(\phi\left(\frac{1}{2}x\right)\right)^2$$

However, no real number squared is -1 and so we've arrived at a contradiction. Therefore, $\mathbf{R}$ under addition is not isomorphic to $\mathbf{R}^*$ under multiplication as desired. $\square$

68. Show that every automorphism ϕ of the rational numbers Q under addition to itself has the form $\phi(x) = x\phi(1)$.

Proof. Let $\phi : Q \rightarrow Q$ be a mapping such that it is an automorphism of the rational numbers Q under addition. Therefore, ϕ is 1-1, onto, and operation-preserving. Furthermore, by **Property 1 of Theorem 6.1**, the identity carries through the mapping, and so

$$\phi(0) = 0$$

For any rational number $x \in Q$, we can write it as $x = \frac{p}{q}$, $p, q \in \mathbb{Z}$ and $q \neq 0$. By applying the automorphism and the property of preserving operations,

$$\phi(x) = \phi\left(\frac{p}{q}\right) = \phi\left(p * \frac{1}{q}\right)$$

Property 2 of Theorem 6.1 states that for every integer n group element $a \in G$, then $\phi(a^n) = [\phi(a)]^n$. Denote this result as *****. Furthermore, in the additive form, $\phi(na) = n\phi(a)$. Denote this result as ******. Therefore, in our context, since $p \in \mathbb{Z}$ and $\frac{1}{q} \in Q$, we can apply ******

$$\phi\left(p * \frac{1}{q}\right) = p\phi\left(\frac{1}{q}\right)$$

Now applying ***** because q is in the group of rationals and -1 is an integer

$$p\phi\left(\frac{1}{q}\right) = p\phi(q^{-1}) = p[\phi(q)]^{-1} = p * \frac{1}{\phi(q)}$$

Applying ****** once more,

$$p * \frac{1}{\phi(q * 1)} = p * \frac{1}{q\phi(1)} = \frac{p}{q}[\phi(1)]^{-1}$$

By ***** for the last time, since -1 is an integer and $1 \in Q$,

$$= \frac{p}{q}\phi(1^{-1}) = \frac{p}{q}\phi\left(\frac{1}{1}\right) = \frac{p}{q}\phi(1) = x\phi(1)$$

and thus we have shown that for any automorphism ϕ of the rational numbers Q under addition, and for any $x \in Q$

$$\phi(x) = x\phi(1)$$

as desired. □